Задание 1

Выбрали ген TP53 (tumor protein p53, ген-супрессор опухолей человека).

Gene ID: 7157

Скачиваем все ортологи белковых последовательностей и скоращаем их количество до 100, чтобы упрстить расчёты.

```
(base) Victoria@DESKTOP-4J9PVD9:~$ datasets download gene symbol TP53 --ortholog all --include protein --filename tp53_orthologs.zip
Collecting 1 gene record [==============================================] 100% 1/1
Collecting 592 gene records [===============================================] 100% 592/592
Collecting 592 gene records [===============================================] 100% 592/592
Downloading: tp53_orthologs.zip    255kB valid data package
Validating package files [===============================================] 100% 5/5
(base) Victoria@DESKTOP-4J9PVD9:~$ unzip tp53_orthologs.zip -d tp53_orthologs
Archive:  tp53_orthologs.zip
  inflating: tp53_orthologs/README.md
  inflating: tp53_orthologs/ncbi_dataset/data/protein.faa
  inflating: tp53_orthologs/ncbi_dataset/data/data_report.jsonl
  inflating: tp53_orthologs/ncbi_dataset/data/dataset_catalog.json
  inflating: tp53_orthologs/md5sum.txt
(base) Victoria@DESKTOP-4J9PVD9:~$
```

```
(base) Victoria@DESKTOP-4J9PVD9:~$ cd ~/tp53_orthologs/ncbi_dataset/data
(base) Victoria@DESKTOP-4J9PVD9:~/tp53_orthologs/ncbi_dataset/data$ awk '/^>/ {count++} count > 100 {exit} {print}' protein.faa > tp53_100_orthologs
.faa
(base) Victoria@DESKTOP-4J9PVD9:~/tp53_orthologs/ncbi_dataset/data$ grep -c "^>" tp53_100_orthologs.faa
100
(base) Victoria@DESKTOP-4J9PVD9:~/tp53_orthologs/ncbi_dataset/data$
```

Переименуем строки заголовков в fasta файле.

```
  GNU nano 8.4                                                    rename_tp53.py
from Bio import SeqIO
import re

input_file = "tp53_100_orthologs.faa"
output_file = "tp53_100_renamed.faa"

with open(output_file, "w") as out_handle:
    for record in SeqIO.parse(input_file, "fasta"):
        header = record.description

        org_match = re.search(r'\[organism=([^\]]+)\]', header)
        acc_match = re.match(r'^(\S+)', header)

        if org_match and acc_match:
            organism = org_match.group(1).strip().replace(" ", "_").replace(".", "_").replace("(", "").replace(")", "")
            accession = acc_match.group(1).strip()
            new_id = f"{organism}_{accession}"

            record.id = new_id
            record.description = ""
        else:
            print(f"Не получилось: {header}")

        SeqIO.write(record, out_handle, "fasta")

print(f"Создан файл: {output_file}")
print(f"Количество последовательностей: ", end="")
```

```
(base) Victoria@DESKTOP-4J9PVD9:~/tp53_orthologs/ncbi_dataset/data$ grep -c "^>" tp53_100_renamed.faa
100
(base) Victoria@DESKTOP-4J9PVD9:~/tp53_orthologs/ncbi_dataset/data$ head -n 20 tp53_100_renamed.faa
>Sus_scrofa_NP_998989.3
MEESQSELGVEPPLSQETFSDLWKLLPENNLLSSELSLAAVNDLLLSPVTNWLDENPDDA
SRVPAPPAATAPAPAAPAPATSWPLSSFVPSQKTYPGSYDFRLGFLHSGTAKSVTCTYSP
ALNKLFCQLAKTCPVQLWVSSPPPPGTRVRAMAIYKKSEYMTEVVRRCPHHERSSDYSDG
LAPPQHLIRVEGNLRAEYLDDRNTFRHSVVVPYEPPEVGSDCTTIHYNFMCNSSCMGGMN
RRPILTIITLEDASGNLLGRNSFEVRVCACPGRDRRTEEENFLKKGQSCPEPPPGSTKRA
LPTSTSSSPVQKKKPLDGEYFTLQIRGRERFEMFRELNDALELKDAQTARESGENRAHSS
HLKSKKGQSPSRHKKPMFKREGPDSD
>Strigops_habroptila_XP_030330235.1
MGSYGVLWGAMGLRCPQFSVVLHKLFCRLAKPCPVQVWVGVPPPPGALLRAVAVYKKSEH
VAEVVRRCPHHERCGGGSDGLAPPQHLIRVEGNPQARYHDDETTKRHSVTVPYEPPEVGS
DCTTVLYNFMCNSSCMGGMNRRPILAILTLEGPAGQVLGRRCFEVRVCACPGRDRRIEEE
NHRRRGGAGGVAKRGEGRGHTVYALPPPVEAPESSKKRVLNPDNEEFSLLVRGRHRYEML
KKINDALEAYEGGAAPERPVKGRRPRGEGLLPRSGKKLLLKGAGPDSD
>Strigops_habroptila_XP_030330237.1
MGSYGVLWGAMGLRCPQFSVVLHKLFCRLAKPCPVQVWVGVPPPPGALLRAVAVYKKSEH
VAEVVRRCPHHERCGGGSDGLAPPQHLIRVEGNPQARYHDDETTKRHSVTVPYEPPEVGS
DCTTVLYNFMCNSSCMGGMNRRPILAILTLEGPAGQVLGRRCFEVRVCACPGRDRRIEEE
NHRRRGGAGGVAKRALPPPVEAPESSKKRVLNPDNEEFSLLVRGRHRYEMLKKINDALEA
YEGGAAPERPVKGRRPRGEGLLPRSGKKLLLKGAGPDSD
(base) Victoria@DESKTOP-4J9PVD9:~/tp53_orthologs/ncbi_dataset/data$
```